

Enabling the Success of DevSecOps with AI

AI impacts DevSecOps in many positive ways — from accelerated testing and better security to quicker delivery of software. It promises to enhance every organisation's potential.



In any organisation with a technology-driven IT department, the pursuit of efficiency, agility and cost-effective high quality software development is an ongoing quest. This desire for rapid software development, aligned with the ever-increasing urgency of business stakeholders, has led to the evolution of modern DevSecOps practices. This article delves into how AI is reshaping the landscape of DevSecOps.

Elevating development and testing with DevSecOps

Successful digital transformation stories have a common denominator — the adoption of modern DevSecOps practices. This transformative process encompasses build-test-

release operations, prioritising speed, quality and value creation.

Development and testing teams now face the challenge of delivering accurate, high quality products that adhere to composable architecture and cloud-based application requirements, along with rapid validation and deployment of the solution. The foundation of this efficiency lies in continuous integration (CI) and continuous delivery (CD). CI standardises development practices, while CD ensures rapid deployment with necessary libraries, configurations and provisioning. These practices, collectively known as DevOps, have proven effective for organisations embracing them.

The testing of developer code at unit level could be integrated within the CI services of most readily available

products, open source or otherwise. However, when it came to functional, system and other types of testing, organisations had built manual or siloed automations. These required significant rework with every change in code along with manual interventions in execution, validation and the reporting aspects of testing. The very nature of separation of duties and the function of testing required a different approach to bring it all together.

Table 1 lists the different types of testing and their roles in any given software development initiative. These testing types are not comprehensive; however, the intent is to showcase the various types of testing involved in a typical application development life cycle and the responsibilities